2 Cauchy's Theorem and Its Applications

> The solution of a large number of problems can be reduced, in the last analysis, to the evaluation of definite integrals; thus mathematicians have been much occupied with this task... However, among many results obtained, a number were initially discovered by the aid of a type of induction based on the passage from real to imaginary. Often passage of this kind led directly to remarkable results. Nevertheless this part of the theory, as has been observed by Laplace, is subject to various difficulties...
>
> After having reflected on this subject and brought together various results mentioned above, I hope to establish the passage from the real to the imaginary based on a direct and rigorous analysis; my researches have thus led me to the method which is the object of this memoir...
>
> *A. L. Cauchy,* 1827

In the previous chapter, we discussed several preliminary ideas in complex analysis: open sets in $\mathbb{C}$, holomorphic functions, and integration along curves. The first remarkable result of the theory exhibits a deep connection between these notions. Loosely stated, Cauchy's theorem says that if f is holomorphic in an open set Ω and $\gamma \subset \Omega$ is a closed curve whose interior is also contained in Ω then

$$\int_\gamma f(z)\, dz = 0. \tag{1}$$

Many results that follow, and in particular the calculus of residues, are related in one way or another to this fact.

A precise and general formulation of Cauchy's theorem requires defining unambiguously the "interior" of a curve, and this is not always an easy task. At this early stage of our study, we shall make use of the device of limiting ourselves to regions whose boundaries are curves that are "toy contours." As the name suggests, these are closed curves whose visualization is so simple that the notion of their interior will be unambiguous, and the proof of Cauchy's theorem in this setting will be quite direct. For many applications, it will suffice to restrict ourselves to these types of curves. At a later stage, we take up the questions related to more general curves, their interiors, and the extended form of Cauchy's theorem.

Our initial version of Cauchy's theorem begins with the observation that it suffices that f have a primitive in Ω, by Corollary 3.3 in Chapter 1. The existence of such a primitive for toy contours will follow from a theorem of Goursat (which is itself a simple special case)[1] that asserts that if f is holomorphic in an open set that contains a triangle T and its interior, then

$$\int_T f(z)\, dz = 0.$$

It is noteworthy that this simple case of Cauchy's theorem suffices to prove some of its more complicated versions. From there, we can prove the existence of primitives in the interior of some simple regions, and therefore prove Cauchy's theorem in that setting. As a first application of this viewpoint, we evaluate several real integrals by using appropriate toy contours.

The above ideas also lead us to a central result of this chapter, the Cauchy integral formula; this states that if f is holomorphic in an open set containing a circle C and its interior, then for all z inside C,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

Differentiation of this identity yields other integral formulas, and in particular we obtain the regularity of holomorphic functions. This is remarkable, since holomorphicity assumed only the existence of the first derivative, and yet we obtain as a consequence the existence of derivatives of all orders. (An analogous statement is decisively false in the case of real variables!)

The theory developed up to that point already has a number of noteworthy consequences:

- The property at the base of "analytic continuation," namely that a holomorphic function is determined by its restriction to any open subset of its domain of definition. This is a consequence of the fact that holomorphic functions have power series expansions.

[1] Goursat's result came after Cauchy's theorem, and its interest is the technical fact that its proof requires only the existence of the complex derivative at each point, and not its continuity. For the earlier proof, see Exercise 5.

- Liouville's theorem, which yields a quick proof of the fundamental theorem of algebra.
- Morera's theorem, which gives a simple integral characterization of holomorphic functions, and shows that these functions are preserved under uniform limits.

1 Goursat's theorem

Corollary 3.3 in the previous chapter says that if f has a primitive in an open set Ω, then

$$\int_\gamma f(z)\,dz = 0$$

for any closed curve γ in Ω. Conversely, if we can show that the above relation holds for some types of curves γ, then a primitive will exist. Our starting point is Goursat's theorem, from which in effect we shall deduce most of the other results in this chapter.

Theorem 1.1 *If Ω is an open set in $\mathbb{C}$, and $T \subset \Omega$ a triangle whose interior is also contained in Ω, then*

$$\int_T f(z)\,dz = 0$$

whenever f is holomorphic in Ω.

Proof. We call $T^{(0)}$ our original triangle (with a fixed orientation which we choose to be positive), and let $d^{(0)}$ and $p^{(0)}$ denote the diameter and perimeter of $T^{(0)}$, respectively. The first step in our construction consists of bisecting each side of the triangle and connecting the midpoints. This creates four new smaller triangles, denoted $T_1^{(1)}, T_2^{(1)}, T_3^{(1)}$, and $T_4^{(1)}$ that are similar to the original triangle. The construction and orientation of each triangle are illustrated in Figure 1. The orientation is chosen to be consistent with that of the original triangle, and so after cancellations arising from integrating over the same side in two opposite directions, we have

(2)

$$\int_{T^{(0)}} f(z)\,dz = \int_{T_1^{(1)}} f(z)\,dz + \int_{T_2^{(1)}} f(z)\,dz + \int_{T_3^{(1)}} f(z)\,dz + \int_{T_4^{(1)}} f(z)\,dz.$$

For some j we must have

$$\left|\int_{T^{(0)}} f(z)\,dz\right| \leq 4\left|\int_{T_j^{(1)}} f(z)\,dz\right|,$$

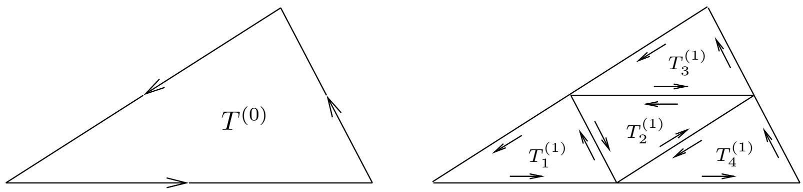

Figure 1. Bisection of $T^{(0)}$

for otherwise (2) would be contradicted. We choose a triangle that satisfies this inequality, and rename it $T^{(1)}$. Observe that if $d^{(1)}$ and $p^{(1)}$ denote the diameter and perimeter of $T^{(1)}$, respectively, then $d^{(1)} = (1/2)d^{(0)}$ and $p^{(1)} = (1/2)p^{(0)}$. We now repeat this process for the triangle $T^{(1)}$, bisecting it into four smaller triangles. Continuing this process, we obtain a sequence of triangles

$$T^{(0)}, T^{(1)}, \ldots, T^{(n)}, \ldots$$

with the properties that

$$\left| \int_{T^{(0)}} f(z)\, dz \right| \leq 4^n \left| \int_{T^{(n)}} f(z)\, dz \right|$$

and

$$d^{(n)} = 2^{-n} d^{(0)}, \qquad p^{(n)} = 2^{-n} p^{(0)}$$

where $d^{(n)}$ and $p^{(n)}$ denote the diameter and perimeter of $T^{(n)}$, respectively. We also denote by $\mathcal{T}^{(n)}$ the *solid* closed triangle with boundary $T^{(n)}$, and observe that our construction yields a sequence of nested compact sets

$$\mathcal{T}^{(0)} \supset \mathcal{T}^{(1)} \supset \cdots \supset \mathcal{T}^{(n)} \supset \cdots$$

whose diameter goes to 0. By Proposition 1.4 in Chapter 1, there exists a unique point z_0 that belongs to all the solid triangles $\mathcal{T}^{(n)}$. Since f is holomorphic at z_0 we can write

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0)\,,$$

where $\psi(z) \to 0$ as $z \to z_0$. Since the constant $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, we can integrate the above equality using Corollary 3.3 in the previous chapter, and obtain

$$(3) \qquad \int_{T^{(n)}} f(z)\, dz = \int_{T^{(n)}} \psi(z)(z - z_0)\, dz.$$

Now z_0 belongs to the closure of the solid triangle $\mathcal{T}^{(n)}$ and z to its boundary, so we must have $|z - z_0| \leq d^{(n)}$, and using (3) we get, by (iii) in Proposition 3.1 of the previous chapter, the estimate

$$\left| \int_{T^{(n)}} f(z)\, dz \right| \leq \epsilon_n d^{(n)} p^{(n)},$$

where $\epsilon_n = \sup_{z \in T^{(n)}} |\psi(z)| \to 0$ as $n \to \infty$. Therefore

$$\left| \int_{T^{(n)}} f(z)\, dz \right| \leq \epsilon_n 4^{-n} d^{(0)} p^{(0)}\,,$$

which yields our final estimate

$$\left| \int_{T^{(0)}} f(z)\, dz \right| \leq 4^n \left| \int_{T^{(n)}} f(z)\, dz \right| \leq \epsilon_n d^{(0)} p^{(0)}.$$

Letting $n \to \infty$ concludes the proof since $\epsilon_n \to 0$.

Corollary 1.2 *If f is holomorphic in an open set Ω that contains a rectangle R and its interior, then*

$$\int_R f(z)\, dz = 0.$$

This is immediate since we first choose an orientation as in Figure 2 and note that

$$\int_R f(z)\, dz = \int_{T_1} f(z)\, dz + \int_{T_2} f(z)\, dz.$$

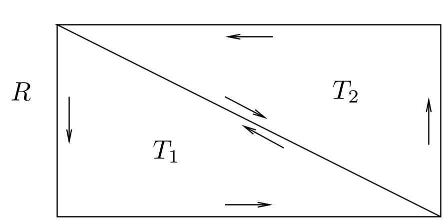

Figure 2. A rectangle as the union of two triangles

2 Local existence of primitives and Cauchy's theorem in a disc

We first prove the existence of primitives in a disc as a consequence of Goursat's theorem.

Theorem 2.1 *A holomorphic function in an open disc has a primitive in that disc.*

Proof. After a translation, we may assume without loss of generality that the disc, say D, is centered at the origin. Given a point $z \in D$, consider the piecewise-smooth curve that joins 0 to z first by moving in the horizontal direction from 0 to $\tilde{z}$ where $\tilde{z} = \mathrm{Re}(z)$, and then in the vertical direction from $\tilde{z}$ to z. We choose the orientation from 0 to z, and denote this polygonal line (which consists of at most two segments) by γ_z, as shown on Figure 3.

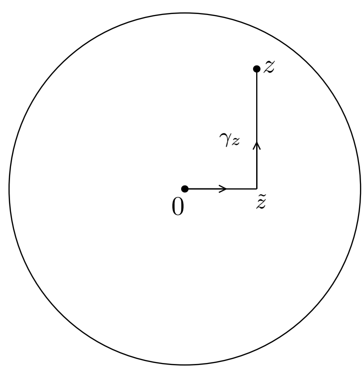

Figure 3. The polygonal line γ_z

Define

$$F(z) = \int_{\gamma_z} f(w)\, dw.$$

The choice of γ_z gives an unambiguous definition of the function $F(z)$. We contend that F is holomorphic in D and $F'(z) = f(z)$. To prove this, fix $z \in D$ and let $h \in \mathbb{C}$ be so small that $z + h$ also belongs to the disc. Now consider the difference

$$F(z+h) - F(z) = \int_{\gamma_{z+h}} f(w)\, dw - \int_{\gamma_z} f(w)\, dw.$$

The function f is first integrated along γ_{z+h} with the original orientation, and then along γ_z with the reverse orientation (because of the minus sign in front of the second integral). This corresponds to (a) in Figure 4. Since we integrate f over the line segment starting at the origin in two opposite directions, it cancels, leaving us with the contour in (b). Then, we complete the square and triangle as shown in (c), so that after an application of Goursat's theorem for triangles and rectangles we are left with the line segment from z to $z+h$ as given in (d).

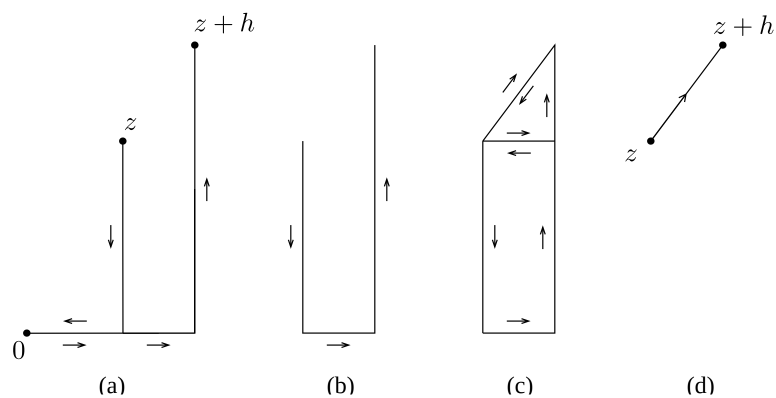

Figure 4. Relation between the polygonal lines γ_z and γ_{z+h}

Hence the above cancellations yield

$$F(z+h) - F(z) = \int_\eta f(w)\, dw$$

where η is the straight line segment from z to $z+h$. Since f is continuous at z we can write

$$f(w) = f(z) + \psi(w)$$

where $\psi(w) \to 0$ as $w \to z$. Therefore

(4)

$$F(z+h) - F(z) = \int_\eta f(z)\, dw + \int_\eta \psi(w)\, dw = f(z) \int_\eta dw + \int_\eta \psi(w)\, dw.$$

On the one hand, the constant 1 has w as a primitive, so the first integral is simply h by an application of Theorem 3.2 in Chapter 1. On the other hand, we have the following estimate:

$$\left|\int_\eta \psi(w)\,dw\right| \leq \sup_{w\in\eta} |\psi(w)|\,|h|.$$

Since the supremum above goes to 0 as h tends to 0, we conclude from equation (4) that

$$\lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = f(z)\,,$$

thereby proving that F is a primitive for f on the disc.

This theorem says that locally, every holomorphic function has a primitive. It is crucial to realize, however, that the theorem is true not only for arbitrary discs, but also for other sets as well. We shall return to this point shortly in our discussion of "toy contours."

Theorem 2.2 (Cauchy's theorem for a disc) *If f is holomorphic in a disc, then*

$$\int_\gamma f(z)\,dz = 0$$

for any closed curve γ in that disc.

Proof. Since f has a primitive, we can apply Corollary 3.3 of Chapter 1.

Corollary 2.3 *Suppose f is holomorphic in an open set containing the circle C and its interior. Then*

$$\int_C f(z)\,dz = 0.$$

Proof. Let D be the disc with boundary circle C. Then there exists a slightly larger disc D' which contains D and so that f is holomorphic on D'. We may now apply Cauchy's theorem in D' to conclude that $\int_C f(z)\,dz = 0$.

In fact, the proofs of the theorem and its corollary apply whenever we can define without ambiguity the "interior" of a contour, and construct appropriate polygonal paths in an open neighborhood of that contour and its interior. In the case of the circle, whose interior is the disc, there was no problem since the geometry of the disc made it simple to travel horizontally and vertically inside it.

The following definition is loosely stated, although its applications will be clear and unambiguous. We call a **toy contour** any closed curve where the notion of interior is obvious, and a construction similar to that in Theorem 2.1 is possible in a neighborhood of the curve and its interior. Its positive orientation is that for which the interior is to the left as we travel along the toy contour. This is consistent with the definition of the positive orientation of a circle. For example, circles, triangles, and rectangles are toy contours, since in each case we can modify (and actually copy) the argument given previously.

Another important example of a toy contour is the "keyhole" Γ (illustrated in Figure 5), which we shall put to use in the proof of the Cauchy integral formula. It consists of two almost complete circles, one large

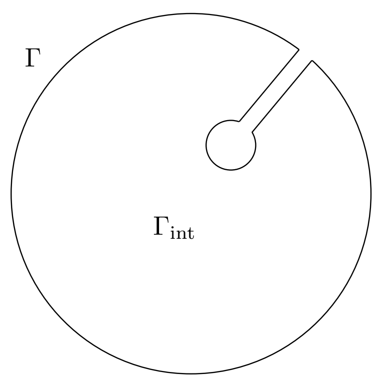

Figure 5. The keyhole contour

and one small, connected by a narrow corridor. The interior of Γ, which we denote by Γ_{int}, is clearly that region enclosed by the curve, and can be given precise meaning with enough work. We fix a point z_0 in that interior. If f is holomorphic in a neighborhood of Γ and its interior, then it is holomorphic in the inside of a slightly larger keyhole, say Λ, whose interior Λ_{int} contains $\Gamma \cup \Gamma_{\text{int}}$. If $z \in \Lambda_{\text{int}}$, let γ_z denote any curve contained inside Λ_{int} connecting z_0 to z, and which consists of finitely many horizontal or vertical segments (as in Figure 6). If η_z is any other such curve, the rectangle version of Goursat's theorem (Corollary 1.2) implies that

$$\int_{\gamma_z} f(w)\, dw = \int_{\eta_z} f(w)\, dw\,,$$

and we may therefore define F unambiguously in Λ_{int}.

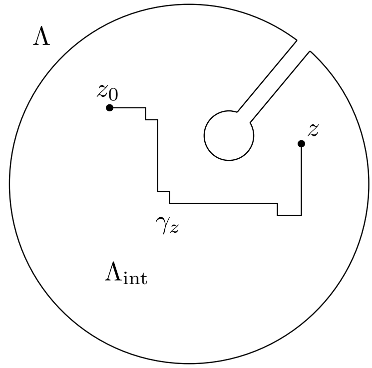

Figure 6. A curve γ_z

Arguing as above allows us to show that F is a primitive of f in Λ_{int} and therefore $\int_\Gamma f(z)\,dz = 0$.

The important point is that for a toy contour γ we easily have that

$$\int_\gamma f(z)\,dz = 0\,,$$

whenever f is holomorphic in an open set that contains the contour γ and its interior.

Other examples of toy contours which we shall encounter in applications and for which Cauchy's theorem and its corollary also hold are given in Figure 7.

While Cauchy's theorem for toy contours is sufficient for most applications we deal with, the question still remains as to what happens for more general curves. We take up this matter in Appendix B, where we prove Jordan's theorem for piecewise-smooth curves. This theorem states that a simple closed piecewise-smooth curve has a well defined interior that is "simply connected." As a consequence, we find that even in this more general situation, Cauchy's theorem holds.

3 Evaluation of some integrals

Here we take up the idea that originally motivated Cauchy. We shall show by several examples how some integrals may be evaluated by the use of his theorem. A more systematic approach, in terms of the calculus of residues, may be found in the next chapter.

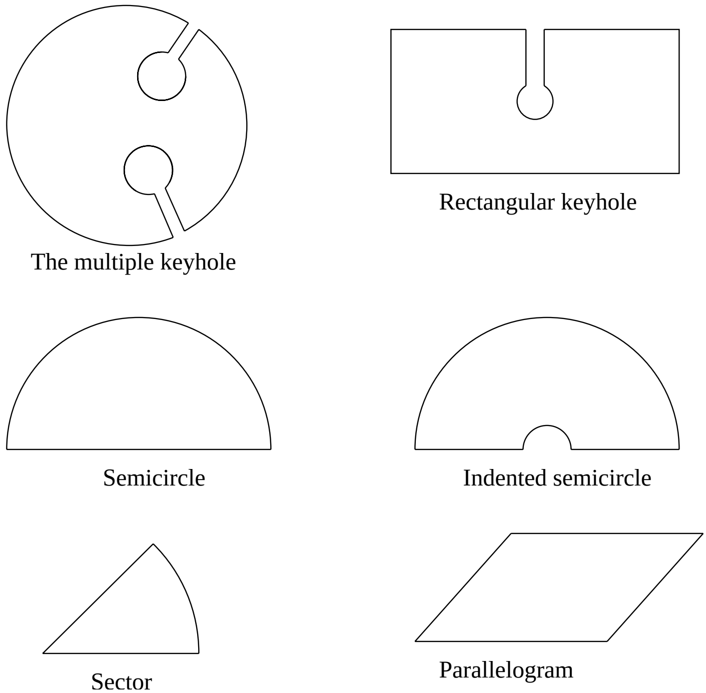

Figure 7. Examples of toy contours

EXAMPLE 1. We show that if $\xi \in \mathbb{R}$, then

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx. \tag{5}$$

This gives a new proof of the fact that $e^{-\pi x^2}$ is its own Fourier transform, a fact we proved in Theorem 1.4 of Chapter 5 in Book I.

If $\xi = 0$, the formula is precisely the known integral[2]

$$1 = \int_{-\infty}^{\infty} e^{-\pi x^2}\, dx.$$

Now suppose that $\xi > 0$, and consider the function $f(z) = e^{-\pi z^2}$, which is entire, and in particular holomorphic in the interior of the toy contour γ_R depicted in Figure 8.

[2] An alternate derivation follows from the fact that $\Gamma(1/2) = \sqrt{\pi}$, where Γ is the gamma function in Chapter 6.

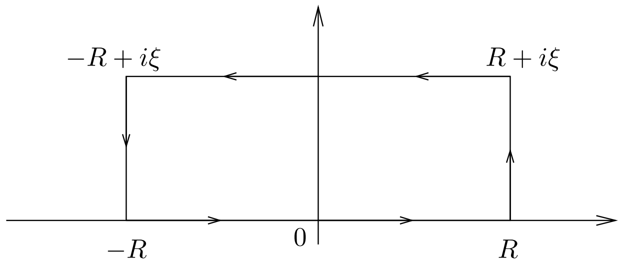

Figure 8. The contour γ_R in Example 1

The contour γ_R consists of a rectangle with vertices $R, R+i\xi, -R+i\xi, -R$ and the positive counterclockwise orientation. By Cauchy's theorem,

$$\int_{\gamma_R} f(z)\, dz = 0. \tag{6}$$

The integral over the real segment is simply

$$\int_{-R}^{R} e^{-\pi x^2}\, dx\,,$$

which converges to 1 as $R \to \infty$. The integral on the vertical side on the right is

$$I(R) = \int_0^\xi f(R+iy) i\, dy = \int_0^\xi e^{-\pi(R^2+2iRy-y^2)} i\, dy.$$

This integral goes to 0 as $R \to \infty$ since ξ is fixed and we may estimate it by

$$|I(R)| \le Ce^{-\pi R^2}.$$

Similarly, the integral over the vertical segment on the left also goes to 0 as $R \to \infty$ for the same reasons. Finally, the integral over the horizontal segment on top is

$$\int_R^{-R} e^{-\pi(x+i\xi)^2}\, dx = -e^{\pi\xi^2} \int_{-R}^{R} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx.$$

Therefore, we find in the limit as $R \to \infty$ that (6) gives

$$0 = 1 - e^{\pi\xi^2} \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx,$$

and our desired formula is established. In the case $\xi < 0$, we then consider the symmetric rectangle, in the lower half-plane.

The technique of shifting the contour of integration, which was used in the previous example, has many other applications. Note that the original integral (5) is taken over the real line, which by an application of Cauchy's theorem is then shifted upwards or downwards (depending on the sign of ξ) in the complex plane.

Example 2. Another classical example is

$$\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}.$$

Here we consider the function $f(z) = (1-e^{iz})/z^2$, and we integrate over the indented semicircle in the upper half-plane positioned on the x-axis, as shown in Figure 9.

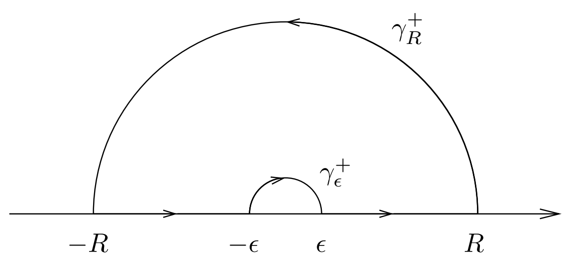

Figure 9. The indented semicircle of Example 2

If we denote by γ_ϵ^+ and γ_R^+ the semicircles of radii ϵ and R with negative and positive orientations respectively, Cauchy's theorem gives

$$\int_{-R}^{-\epsilon} \frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_\epsilon^+} \frac{1-e^{iz}}{z^2}\,dz + \int_\epsilon^R \frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_R^+} \frac{1-e^{iz}}{z^2}\,dz = 0.$$

First we let $R \to \infty$ and observe that

$$\left|\frac{1-e^{iz}}{z^2}\right| \le \frac{2}{|z|^2},$$

so the integral over γ_R^+ goes to zero. Therefore

$$\int_{|x|\ge\epsilon} \frac{1-e^{ix}}{x^2}\,dx = -\int_{\gamma_\epsilon^+} \frac{1-e^{iz}}{z^2}\,dz.$$

Next, note that

$$f(z) = \frac{-iz}{z^2} + E(z)$$

where $E(z)$ is bounded as $z \to 0$, while on γ_ϵ^+ we have $z = \epsilon e^{i\theta}$ and $dz = i\epsilon e^{i\theta} d\theta$. Thus

$$\int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2}\, dz \to \int_\pi^0 (-ii)\, d\theta = -\pi \qquad \text{as } \epsilon \to 0.$$

Taking real parts then yields

$$\int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2}\, dx = \pi.$$

Since the integrand is even, the desired formula is proved.

4 Cauchy's integral formulas

Representation formulas, and in particular integral representation formulas, play an important role in mathematics, since they allow us to recover a function on a large set from its behavior on a smaller set. For example, we saw in Book I that a solution of the steady-state heat equation in the disc was completely determined by its boundary values on the circle via a convolution with the Poisson kernel

$$u(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(1, \varphi)\, d\varphi. \tag{7}$$

In the case of holomorphic functions, the situation is analogous, which is not surprising since the real and imaginary parts of a holomorphic function are harmonic.[3] Here, we will prove an integral representation formula in a manner that is independent of the theory of harmonic functions. In fact, it is also possible to recover the Poisson integral formula (7) as a consequence of the next theorem (see Exercises 11 and 12).

Theorem 4.1 *Suppose f is holomorphic in an open set that contains the closure of a disc D. If C denotes the boundary circle of this disc with the positive orientation, then*

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \textit{for any point } z \in D.$$

[3] This fact is an immediate consequence of the Cauchy-Riemann equations. We refer the reader to Exercise 11 in Chapter 1.

Proof. Fix $z \in D$ and consider the "keyhole" $\Gamma_{\delta,\epsilon}$ which omits the point z as shown in Figure 10.

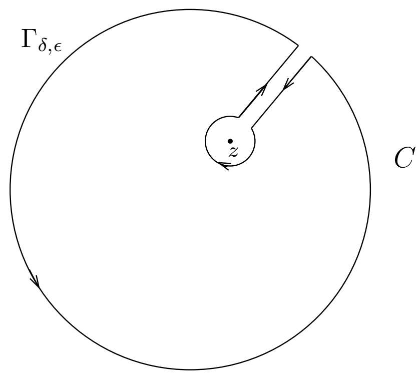

Figure 10. The keyhole $\Gamma_{\delta,\epsilon}$

Here δ is the width of the corridor, and ϵ the radius of the small circle centered at z. Since the function $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic away from the point $\zeta = z$, we have

$$\int_{\Gamma_{\delta,\epsilon}} F(\zeta)\, d\zeta = 0$$

by Cauchy's theorem for the chosen toy contour. Now we make the corridor narrower by letting δ tend to 0, and use the continuity of F to see that in the limit, the integrals over the two sides of the corridor cancel out. The remaining part consists of two curves, the large boundary circle C with the positive orientation, and a small circle C_ϵ centered at z of radius ϵ and oriented negatively, that is, clockwise. To see what happens to the integral over the small circle we write

$$F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z} \tag{8}$$

and note that since f is holomorphic the first term on the right-hand side of (8) is bounded so that its integral over C_ϵ goes to 0 as $\epsilon \to 0$. To conclude the proof, it suffices to observe that

$$\begin{aligned}\int_{C_\epsilon} \frac{f(z)}{\zeta - z}\, d\zeta &= f(z) \int_{C_\epsilon} \frac{d\zeta}{\zeta - z} \\ &= -f(z) \int_0^{2\pi} \frac{\epsilon i e^{-it}}{\epsilon e^{-it}}\, dt \\ &= -f(z) 2\pi i\,,\end{aligned}$$

so that in the limit we find

$$0 = \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta - 2\pi i f(z)\,,$$

as was to be shown.

Remarks. Our earlier discussion of toy contours provides simple extensions of the Cauchy integral formula; for instance, if f is holomorphic in an open set that contains a (positively oriented) rectangle R and its interior, then

$$f(z) = \frac{1}{2\pi i} \int_R \frac{f(\zeta)}{\zeta - z}\, d\zeta\,,$$

whenever z belongs to the interior of R. To establish this result, it suffices to repeat the proof of Theorem 4.1 replacing the "circular" keyhole by a "rectangular" keyhole.

It should also be noted that the above integral vanishes when z is outside R, since in this case $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic inside R. Of course, a similar result also holds for the circle or any other toy contour.

As a corollary to the Cauchy integral formula, we arrive at a second remarkable fact about holomorphic functions, namely their regularity. We also obtain further integral formulas expressing the derivatives of f inside the disc in terms of the values of f on the boundary.

Corollary 4.2 *If f is holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then*

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta$$

for all z in the interior of C.

We recall that, as in the above theorem, we take the circle C to have positive orientation.

Proof. The proof is by induction on n, the case $n=0$ being simply the Cauchy integral formula. Suppose that f has up to $n-1$ complex derivatives and that

$$f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^n}\, d\zeta.$$

Now for h small, the difference quotient for $f^{(n-1)}$ takes the form

$$\frac{f^{(n-1)}(z+h) - f^{(n-1)}(z)}{h} = \tag{9}$$

$$\frac{(n-1)!}{2\pi i}\int_C f(\zeta)\frac{1}{h}\left[\frac{1}{(\zeta - z - h)^n} - \frac{1}{(\zeta - z)^n}\right] d\zeta.$$

We now recall that

$$A^n - B^n = (A-B)[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

With $A = 1/(\zeta - z - h)$ and $B = 1/(\zeta - z)$, we see that the term in brackets in equation (9) is equal to

$$\frac{h}{(\zeta - z - h)(\zeta - z)}[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

But observe that if h is small, then $z+h$ and z stay at a finite distance from the boundary circle C, so in the limit as h tends to 0, we find that the quotient converges to

$$\frac{(n-1)!}{2\pi i}\int_C f(\zeta)\left[\frac{1}{(\zeta - z)^2}\right]\left[\frac{n}{(\zeta - z)^{n-1}}\right] d\zeta = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta\,,$$

which completes the induction argument and proves the theorem.

From now on, we call the formulas of Theorem 4.1 and Corollary 4.2 the **Cauchy integral formulas**.

Corollary 4.3 (Cauchy inequalities) *If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 and of radius R, then*

$$|f^{(n)}(z_0)| \leq \frac{n!\|f\|_C}{R^n}\,,$$

where $\|f\|_C = \sup_{z \in C}|f(z)|$ denotes the supremum of $|f|$ on the boundary circle C.

Proof. Applying the Cauchy integral formula for $f^{(n)}(z_0)$, we obtain

$$\begin{aligned}
|f^{(n)}(z_0)| &= \left| \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta \right| \\
&= \frac{n!}{2\pi} \left| \int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Rie^{i\theta}\, d\theta \right| \\
&\leq \frac{n!}{2\pi} \frac{\|f\|_C}{R^n} 2\pi.
\end{aligned}$$

Another striking consequence of the Cauchy integral formula is its connection with power series. In Chapter 1, we proved that a power series is holomorphic in the interior of its disc of convergence, and promised a proof of a converse, which is the content of the next theorem.

Theorem 4.4 *Suppose f is holomorphic in an open set Ω. If D is a disc centered at z_0 and whose closure is contained in Ω, then f has a power series expansion at z_0*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D$, and the coefficients are given by

$$a_n = \frac{f^{(n)}(z_0)}{n!} \qquad \text{for all } n \geq 0.$$

Proof. Fix $z \in D$. By the Cauchy integral formula, we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta\,, \tag{10}$$

where C denotes the boundary of the disc and $z \in D$. The idea is to write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0} \frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)}, \tag{11}$$

and use the geometric series expansion. Since $\zeta \in C$ and $z \in D$ is fixed, there exists $0 < r < 1$ such that

$$\left| \frac{z - z_0}{\zeta - z_0} \right| < r,$$

therefore

$$\frac{1}{1-\left(\frac{z-z_0}{\zeta-z_0}\right)} = \sum_{n=0}^{\infty}\left(\frac{z-z_0}{\zeta-z_0}\right)^n, \tag{12}$$

where the series converges uniformly for $\zeta \in C$. This allows us to interchange the infinite sum with the integral when we combine (10), (11), and (12), thereby obtaining

$$f(z) = \sum_{n=0}^{\infty}\left(\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta-z_0)^{n+1}}\,d\zeta\right)\cdot(z-z_0)^n.$$

This proves the power series expansion; further the use of the Cauchy integral formulas for the derivatives (or simply differentiation of the series) proves the formula for a_n.

Observe that since power series define indefinitely (complex) differentiable functions, the theorem gives another proof that a holomorphic function is automatically indefinitely differentiable.

Another important observation is that the power series expansion of f centered at z_0 converges in any disc, no matter how large, as long as its closure is contained in Ω. In particular, if f is entire (that is, holomorphic on all of $\mathbb{C}$), the theorem implies that f has a power series expansion around 0, say $f(z) = \sum_{n=0}^{\infty} a_n z^n$, that converges in all of $\mathbb{C}$.

Corollary 4.5 (Liouville's theorem) *If f is entire and bounded, then f is constant.*

Proof. It suffices to prove that $f' = 0$, since $\mathbb{C}$ is connected, and we may then apply Corollary 3.4 in Chapter 1.

For each $z_0 \in \mathbb{C}$ and all $R > 0$, the Cauchy inequalities yield

$$|f'(z_0)| \leq \frac{B}{R}$$

where B is a bound for f. Letting $R \to \infty$ gives the desired result.

As an application of our work so far, we can give an elegant proof of the fundamental theorem of algebra.

Corollary 4.6 *Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_0$ with complex coefficients has a root in $\mathbb{C}$.*

Proof. If P has no roots, then $1/P(z)$ is a bounded holomorphic function. To see this, we can of course assume that $a_n \neq 0$, and write

$$\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n}\right)$$

whenever $z \neq 0$. Since each term in the parentheses goes to 0 as $|z| \to \infty$ we conclude that there exists $R > 0$ so that if $c = |a_n|/2$, then

$$|P(z)| \geq c|z|^n \qquad \text{whenever } |z| > R.$$

In particular, P is bounded from below when $|z| > R$. Since P is continuous and has no roots in the disc $|z| \leq R$, it is bounded from below in that disc as well, thereby proving our claim.

By Liouville's theorem we then conclude that $1/P$ is constant. This contradicts our assumption that P is non-constant and proves the corollary.

Corollary 4.7 *Every polynomial* $P(z) = a_n z^n + \cdots + a_0$ *of degree* $n \geq 1$ *has precisely* n *roots in* $\mathbb{C}$. *If these roots are denoted by* $w_1, \ldots, w_n$, *then* P *can be factored as*

$$P(z) = a_n(z - w_1)(z - w_2)\cdots(z - w_n).$$

Proof. By the previous result P has a root, say w_1. Then, writing $z = (z - w_1) + w_1$, inserting this expression for z in P, and using the binomial formula we get

$$P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0,$$

where $b_0, \ldots, b_{n-1}$ are new coefficients, and $b_n = a_n$. Since $P(w_1) = 0$, we find that $b_0 = 0$, therefore

$$P(z) = (z - w_1)\left[b_n(z - w_1)^{n-1} + \cdots + b_1\right] = (z - w_1)Q(z),$$

where Q is a polynomial of degree $n - 1$. By induction on the degree of the polynomial, we conclude that $P(z)$ has n roots and can be expressed as

$$P(z) = c(z - w_1)(z - w_2)\cdots(z - w_n)$$

for some $c \in \mathbb{C}$. Expanding the right-hand side, we realize that the coefficient of z^n is c and therefore $c = a_n$ as claimed.

Finally, we end this section with a discussion of analytic continuation (the third of the "miracles" we mentioned in the introduction). It states that the "genetic code" of a holomorphic function is determined (that is, the function is fixed) if we know its values on appropriate arbitrarily small subsets. Note that in the theorem below, Ω is assumed connected.

Theorem 4.8 *Suppose f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω. Then f is identically* 0.

In other words, if the zeros of a holomorphic function f in the connected open set Ω accumulate *in* Ω, then $f = 0$.

Proof. Suppose that $z_0 \in \Omega$ is a limit point for the sequence $\{w_k\}_{k=1}^{\infty}$ and that $f(w_k) = 0$. First, we show that f is identically zero in a small disc containing z_0. For that, we choose a disc D centered at z_0 and contained in Ω, and consider the power series expansion of f in that disc

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

If f is not identically zero, there exists a smallest integer m such that $a_m \neq 0$. But then we can write

$$f(z) = a_m (z - z_0)^m (1 + g(z - z_0)),$$

where $g(z - z_0)$ converges to 0 as $z \to z_0$. Taking $z = w_k \neq z_0$ for a sequence of points converging to z_0, we get a contradiction since $a_m (w_k - z_0)^m \neq 0$ and $1 + g(w_k - z_0) \neq 0$, but $f(w_k) = 0$.

We conclude the proof using the fact that Ω is connected. Let U denote the interior of the set of points where $f(z) = 0$. Then U is open by definition and non-empty by the argument just given. The set U is also closed since if $z_n \in U$ and $z_n \to z$, then $f(z) = 0$ by continuity, and f vanishes in a neighborhood of z by the argument above. Hence $z \in U$. Now if we let V denote the complement of U in Ω, we conclude that U and V are both open, disjoint, and

$$\Omega = U \cup V.$$

Since Ω is connected we conclude that either U or V is empty. (Here we use one of the two equivalent definitions of connectedness discussed in Chapter 1.) Since $z_0 \in U$, we find that $U = \Omega$ and the proof is complete.

An immediate consequence of the theorem is the following.

Corollary 4.9 *Suppose f and g are holomorphic in a region Ω and $f(z) = g(z)$ for all z in some non-empty open subset of Ω (or more generally for z in some sequence of distinct points with limit point in Ω). Then $f(z) = g(z)$ throughout Ω.*

Suppose we are given a pair of functions f and F analytic in regions Ω and Ω', respectively, with $\Omega \subset \Omega'$. If the two functions agree on the smaller set Ω, we say that F is an **analytic continuation** of f into the region Ω'. The corollary then guarantees that there can be only one such analytic continuation, since F is uniquely determined by f.

5 Further applications

We gather in this section various consequences of the results proved so far.

5.1 Morera's theorem

A direct application of what was proved here is a converse of Cauchy's theorem.

Theorem 5.1 *Suppose f is a continuous function in the open disc D such that for any triangle T contained in D*

$$\int_T f(z)\, dz = 0,$$

then f is holomorphic.

Proof. By the proof of Theorem 2.1 the function f has a primitive F in D that satisfies $F' = f$. By the regularity theorem, we know that F is indefinitely (and hence twice) complex differentiable, and therefore f is holomorphic.

5.2 Sequences of holomorphic functions

Theorem 5.2 *If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.*

Proof. Let D be any disc whose closure is contained in Ω and T any triangle in that disc. Then, since each f_n is holomorphic, Goursat's theorem implies

$$\int_T f_n(z)\, dz = 0 \qquad \text{for all } n.$$

By assumption $f_n \to f$ uniformly in the closure of D, so f is continuous and

$$\int_T f_n(z)\, dz \to \int_T f(z)\, dz.$$

As a result, we find $\int_T f(z)\,dz = 0$, and by Morera's theorem, we conclude that f is holomorphic in D. Since this conclusion is true for every D whose closure is contained in Ω, we find that f is holomorphic in all of Ω.

This is a striking result that is obviously not true in the case of real variables: the uniform limit of continuously differentiable functions need not be differentiable. For example, we know that every continuous function on $[0, 1]$ can be approximated uniformly by polynomials, by Weierstrass's theorem (see Chapter 5, Book I), yet not every continuous function is differentiable.

We can go one step further and deduce convergence theorems for the sequence of derivatives. Recall that if f is a power series with radius of convergence R, then f' can be obtained by differentiating term by term the series for f, and moreover f' has radius of convergence R. (See Theorem 2.6 in Chapter 1.) This implies in particular that if S_n are the partial sums of f, then S_n' converges uniformly to f' on every compact subset of the disc of convergence of f. The next theorem generalizes this fact.

Theorem 5.3 *Under the hypotheses of the previous theorem, the sequence of derivatives $\{f_n'\}_{n=1}^{\infty}$ converges uniformly to f' on every compact subset of Ω.*

Proof. We may assume without loss of generality that the sequence of functions in the theorem converges uniformly on all of Ω. Given $\delta > 0$, let Ω_δ denote the subset of Ω defined by

$$\Omega_\delta = \{z \in \Omega : \overline{D_\delta(z)} \subset \Omega\}.$$

In other words, Ω_δ consists of all points in Ω which are at distance $> \delta$ from its boundary. To prove the theorem, it suffices to show that $\{f_n'\}$ converges uniformly to f' on Ω_δ for each δ. This is achieved by proving the following inequality:

$$\sup_{z\in\Omega_\delta} |F'(z)| \le \frac{1}{\delta} \sup_{\zeta\in\Omega} |F(\zeta)| \tag{13}$$

whenever F is holomorphic in Ω, since it can then be applied to $F = f_n - f$ to prove the desired fact. The inequality (13) follows at once from the Cauchy integral formula and the definition of Ω_δ, since for every $z \in \Omega_\delta$ the closure of $D_\delta(z)$ is contained in Ω and

$$F'(z) = \frac{1}{2\pi i} \int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2}\, d\zeta.$$

Hence,

$$
\begin{aligned}
|F'(z)| &\leq \frac{1}{2\pi} \int_{C_\delta(z)} \frac{|F(\zeta)|}{|\zeta - z|^2} \, |d\zeta| \\
&\leq \frac{1}{2\pi} \sup_{\zeta \in \Omega} |F(\zeta)| \, \frac{1}{\delta^2} \, 2\pi\delta \\
&= \frac{1}{\delta} \sup_{\zeta \in \Omega} |F(\zeta)| \, ,
\end{aligned}
$$

as was to be shown.

Of course, there is nothing special about the first derivative, and in fact under the hypotheses of the last theorem, we may conclude (arguing as above) that for every $k \geq 0$ the sequence of k^{th} derivatives $\{f_n^{(k)}\}_{n=1}^\infty$ converges uniformly to $f^{(k)}$ on every compact subset of Ω.

In practice, one often uses Theorem 5.2 to construct holomorphic functions (say, with a prescribed property) as a series

$$F(z) = \sum_{n=1}^{\infty} f_n(z). \tag{14}$$

Indeed, if each f_n is holomorphic in a given region Ω of the complex plane, and the series converges uniformly in compact subsets of Ω, then Theorem 5.2 guarantees that F is also holomorphic in Ω. For instance, various special functions are often expressed in terms of a converging series like (14). A specific example is the Riemann zeta function discussed in Chapter 6.

We now turn to a variant of this idea, which consists of functions defined in terms of integrals.

5.3 Holomorphic functions defined in terms of integrals

As we shall see later in this book, a number of other special functions are defined in terms of integrals of the type

$$f(z) = \int_a^b F(z, s) \, ds,$$

or as limits of such integrals. Here, the function F is holomorphic in the first argument, and continuous in the second. The integral is taken in the sense of Riemann integration over the bounded interval $[a, b]$. The problem then is to establish that f is holomorphic.

In the next theorem, we impose a sufficient condition on F, often satisfied in practice, that easily implies that f is holomorphic.

After a simple linear change of variables, we may assume that $a = 0$ and $b = 1$.

Theorem 5.4 *Let $F(z, s)$ be defined for $(z, s) \in \Omega \times [0, 1]$ where Ω is an open set in $\mathbb{C}$. Suppose F satisfies the following properties:*

(i) *$F(z, s)$ is holomorphic in z for each s.*

(ii) *F is continuous on $\Omega \times [0, 1]$.*

Then the function f defined on Ω by

$$f(z) = \int_0^1 F(z, s)\, ds$$

is holomorphic.

The second condition says that F is jointly continuous in both arguments.

To prove this result, it suffices to prove that f is holomorphic in any disc D contained in Ω, and by Morera's theorem this could be achieved by showing that for any triangle T contained in D we have

$$\int_T \int_0^1 F(z, s)\, ds\, dz = 0.$$

Interchanging the order of integration, and using property (i) would then yield the desired result. We can, however, get around the issue of justifying the change in the order of integration by arguing differently. The idea is to interpret the integral as a "uniform" limit of Riemann sums, and then apply the results of the previous section.

Proof. For each $n \geq 1$, we consider the Riemann sum

$$f_n(z) = (1/n) \sum_{k=1}^{n} F(z, k/n).$$

Then f_n is holomorphic in all of Ω by property (i), and we claim that on any disc D whose closure is contained in Ω, the sequence $\{f_n\}_{n=1}^{\infty}$ converges uniformly to f. To see this, we recall that a continuous function on a compact set is uniformly continuous, so if $\epsilon > 0$ there exists $\delta > 0$ such that

$$\sup_{z \in D} |F(z, s_1) - F(z, s_2)| < \epsilon \quad \text{whenever } |s_1 - s_2| < \delta.$$

Then, if $n > 1/\delta$, and $z \in D$ we have

$$\begin{aligned} |f_n(z) - f(z)| &= \left| \sum_{k=1}^{n} \int_{(k-1)/n}^{k/n} F(z, k/n) - F(z, s)\, ds \right| \\ &\leq \sum_{k=1}^{n} \int_{(k-1)/n}^{k/n} |F(z, k/n) - F(z, s)|\, ds \\ &< \sum_{k=1}^{n} \frac{\epsilon}{n} \\ &< \epsilon. \end{aligned}$$

This proves the claim, and by Theorem 5.2 we conclude that f is holomorphic in D. As a consequence, f is holomorphic in Ω, as was to be shown.

5.4 Schwarz reflection principle

In real analysis, there are various situations where one wishes to extend a function from a given set to a larger one. Several techniques exist that provide extensions for continuous functions, and more generally for functions with varying degrees of smoothness. Of course, the difficulty of the technique increases as we impose more conditions on the extension.

The situation is very different for holomorphic functions. Not only are these functions indefinitely differentiable in their domain of definition, but they also have additional characteristically rigid properties, which make them difficult to mold. For example, there exist holomorphic functions in a disc which are continuous on the closure of the disc, but which cannot be continued (analytically) into any region larger than the disc. (This phenomenon is discussed in Problem 1.) Another fact we have seen above is that holomorphic functions must be identically zero if they vanish on small open sets (or even, for example, a non-zero line segment).

It turns out that the theory developed in this chapter provides a simple extension phenomenon that is very useful in applications: the Schwarz reflection principle. The proof consists of two parts. First we define the extension, and then check that the resulting function is still holomorphic. We begin with this second point.

Let Ω be an open subset of $\mathbb{C}$ that is symmetric with respect to the real line, that is

$$z \in \Omega \quad \text{if and only if} \quad \overline{z} \in \Omega.$$

Let Ω^+ denote the part of Ω that lies in the upper half-plane and Ω^- that part that lies in the lower half-plane.

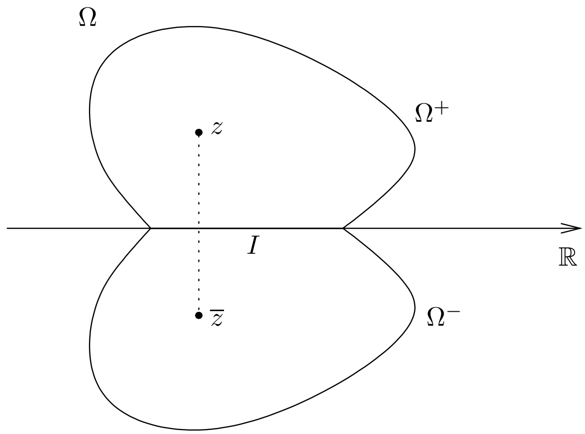

Figure 11. An open set symmetric across the real axis

Also, let $I = \Omega \cap \mathbb{R}$ so that I denotes the interior of that part of the boundary of Ω^+ and Ω^- that lies on the real axis. Then we have

$$\Omega^+ \cup I \cup \Omega^- = \Omega$$

and the only interesting case of the next theorem occurs, of course, when I is non-empty.

Theorem 5.5 (Symmetry principle) *If f^+ and f^- are holomorphic functions in Ω^+ and Ω^- respectively, that extend continuously to I and*

$$f^+(x) = f^-(x) \quad \text{for all } x \in I,$$

then the function f defined on Ω by

$$f(z) = \begin{cases} f^+(z) & \text{if } z \in \Omega^+, \\ f^+(z) = f^-(z) & \text{if } z \in I, \\ f^-(z) & \text{if } z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω.

Proof. One notes first that f is continuous throughout Ω. The only difficulty is to prove that f is holomorphic at points of I. Suppose D is a disc centered at a point on I and entirely contained in Ω. We prove that f is holomorphic in D by Morera's theorem. Suppose T is a triangle in D. If T does not intersect I, then

$$\int_T f(z)\,dz = 0$$

since f is holomorphic in the upper and lower half-discs. Suppose now that one side or vertex of T is contained in I, and the rest of T is in, say, the upper half-disc. If T_ϵ is the triangle obtained from T by slightly raising the edge or vertex which lies on I, we have $\int_{T_\epsilon} f = 0$ since T_ϵ is entirely contained in the upper half-disc (an illustration of the case when an edge lies on I is given in Figure 12(a)). We then let $\epsilon \to 0$, and by continuity we conclude that

$$\int_T f(z)\,dz = 0.$$

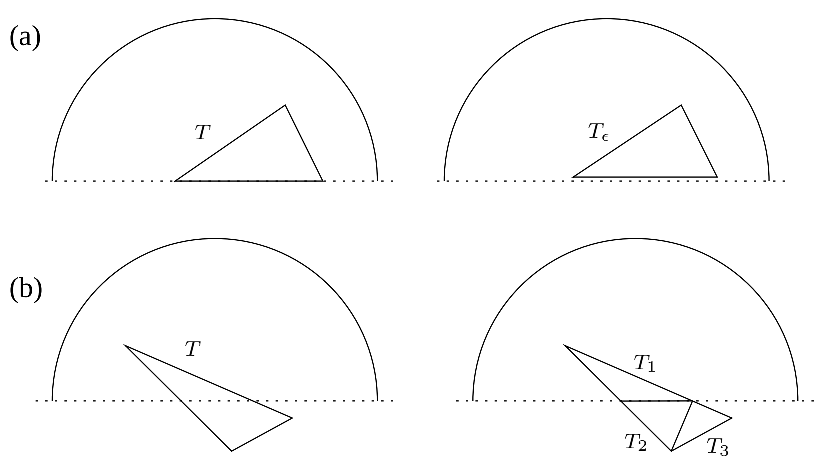

Figure 12. (a) Raising a vertex; (b) splitting a triangle

If the interior of T intersects I, we can reduce the situation to the previous one by writing T as the union of triangles each of which has an edge or vertex on I as shown in Figure 12(b). By Morera's theorem we conclude that f is holomorphic in D, as was to be shown.

We can now state the extension principle, where we use the above notation.

Theorem 5.6 (Schwarz reflection principle) *Suppose that f is a holomorphic function in Ω^+ that extends continuously to I and such that f is real-valued on I. Then there exists a function F holomorphic in all of Ω such that $F = f$ on Ω^+.*

Proof. The idea is simply to define $F(z)$ for $z \in \Omega^-$ by

$$F(z) = \overline{f(\overline{z})}.$$

To prove that F is holomorphic in Ω^- we note that if $z, z_0 \in \Omega^-$, then $\overline{z}, \overline{z_0} \in \Omega^+$ and hence, the power series expansion of f near $\overline{z_0}$ gives

$$f(\overline{z}) = \sum a_n(\overline{z} - \overline{z_0})^n.$$

As a consequence we see that

$$F(z) = \sum \overline{a_n}(z - z_0)^n$$

and F is holomorphic in Ω^-. Since f is real valued on I we have $\overline{f(x)} = f(x)$ whenever $x \in I$ and hence F extends continuously up to I. The proof is complete once we invoke the symmetry principle.

5.5 Runge's approximation theorem

We know by Weierstrass's theorem that any continuous function on a compact interval can be approximated uniformly by polynomials.[4] With this result in mind, one may inquire about similar approximations in complex analysis. More precisely, we ask the following question: what conditions on a compact set $K \subset \mathbb{C}$ guarantee that any function holomorphic in a neighborhood of this set can be approximated uniformly by polynomials on K?

An example of this is provided by power series expansions. We recall that if f is a holomorphic function in a disc D, then it has a power series expansion $f(z) = \sum_{n=0}^{\infty} a_n z^n$ that converges uniformly on every compact set $K \subset D$. By taking partial sums of this series, we conclude that f can be approximated uniformly by polynomials on any compact subset of D.

In general, however, some condition on K must be imposed, as we see by considering the function $f(z) = 1/z$ on the unit circle $K = C$. Indeed, recall that $\int_C f(z)\,dz = 2\pi i$, and if p is any polynomial, then Cauchy's theorem implies $\int_C p(z)\,dz = 0$, and this quickly leads to a contradiction.

[4] A proof may be found in Section 1.8, Chapter 5, of Book I.

A restriction on K that guarantees the approximation pertains to the topology of its complement: K^c must be connected. In fact, a slight modification of the above example when $f(z) = 1/z$ proves that this condition on K is also necessary; see Problem 4.

Conversely, uniform approximations exist when K^c is connected, and this result follows from a theorem of Runge which states that for *any* K a uniform approximation exists by *rational functions* with "singularities" in the complement of K.[5] This result is remarkable since rational functions are globally defined, while f is given only in a neighborhood of K. In particular, f could be defined independently on different components of K, making the conclusion of the theorem even more striking.

Theorem 5.7 *Any function holomorphic in a neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^c.*

If K^c is connected, any function holomorphic in a neighborhood of K can be approximated uniformly on K by polynomials.

We shall see how the second part of the theorem follows from the first: when K^c is connected, one can "push" the singularities to infinity thereby transforming the rational functions into polynomials.

The key to the theorem lies in an integral representation formula that is a simple consequence of the Cauchy integral formula applied to a square.

Lemma 5.8 *Suppose f is holomorphic in an open set Ω, and $K \subset \Omega$ is compact. Then, there exists finitely many segments $\gamma_1, \ldots, \gamma_N$ in $\Omega - K$ such that*

$$f(z) = \sum_{n=1}^{N} \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z}\, d\zeta \quad \textit{for all } z \in K. \tag{15}$$

Proof. Let $d = c \cdot d(K, \Omega^c)$, where c is any constant $< 1/\sqrt{2}$, and consider a grid formed by (solid) squares with sides parallel to the axis and of length d.

We let $\mathcal{Q} = \{Q_1, \ldots, Q_M\}$ denote the finite collection of squares in this grid that intersect K, with the boundary of each square given the positive orientation. (We denote by ∂Q_m the boundary of the square Q_m.) Finally, we let $\gamma_1, \ldots, \gamma_N$ denote the sides of squares in $\mathcal{Q}$ that do not belong to two adjacent squares in $\mathcal{Q}$. (See Figure 13.) The choice of d guarantees that for each n, $\gamma_n \subset \Omega$, and γ_n does not intersect K; for if it did, then it would belong to two adjacent squares in $\mathcal{Q}$, contradicting our choice of γ_n.

[5] These singularities are points where the function is not holomorphic, and are "poles", as defined in the next chapter.

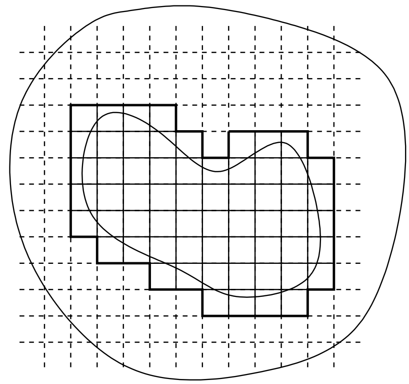

Figure 13. The union of the γ_n's is in bold-face

Since for any $z \in K$ that is not on the boundary of a square in $\mathcal{Q}$ there exists j so that $z \in Q_j$, Cauchy's theorem implies

$$\frac{1}{2\pi i}\int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \begin{cases} f(z) & \text{if } m = j, \\ 0 & \text{if } m \neq j. \end{cases}$$

Thus, for all such z we have

$$f(z) = \sum_{m=1}^{M} \frac{1}{2\pi i}\int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

However, if Q_m and $Q_{m'}$ are adjacent, the integral over their common side is taken once in each direction, and these cancel. This establishes (15) when z is in K and not on the boundary of a square in $\mathcal{Q}$. Since $\gamma_n \subset K^c$, continuity guarantees that this relation continues to hold for all $z \in K$, as was to be shown.

The first part of Theorem 5.7 is therefore a consequence of the next lemma.

Lemma 5.9 *For any line segment γ entirely contained in $\Omega - K$, there exists a sequence of rational functions with singularities on γ that approximate the integral $\int_\gamma f(\zeta)/(\zeta - z)\, d\zeta$ uniformly on K.*

Proof. If $\gamma(t) : [0,1] \to \mathbb{C}$ is a parametrization for γ, then

$$\int_\gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta = \int_0^1 \frac{f(\gamma(t))}{\gamma(t) - z}\gamma'(t)\, dt.$$

Since γ does not intersect K, the integrand $F(z,t)$ in this last integral is jointly continuous on $K \times [0,1]$, and since K is compact, given $\epsilon > 0$, there exists $\delta > 0$ such that

$$\sup_{z \in K} |F(z,t_1) - F(z,t_2)| < \epsilon \quad \text{whenever } |t_1 - t_2| < \delta.$$

Arguing as in the proof of Theorem 5.4, we see that the Riemann sums of the integral $\int_0^1 F(z,t)\,dt$ approximate it uniformly on K. Since each of these Riemann sums is a rational function with singularities on γ, the lemma is proved.

Finally, the process of pushing the poles to infinity is accomplished by using the fact that K^c is connected. Since any rational function whose only singularity is at the point z_0 is a polynomial in $1/(z - z_0)$, it suffices to establish the next lemma to complete the proof of Theorem 5.7.

Lemma 5.10 *If K^c is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.*

Proof. First, we choose a point z_1 that is outside a large open disc D centered at the origin and which contains K. Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1}\frac{1}{1 - z/z_1} = \sum_{n=1}^{\infty} -\frac{z^n}{z_1^{n+1}},$$

where the series converges uniformly for $z \in K$. The partial sums of this series are polynomials that provide a uniform approximation to $1/(z - z_1)$ on K. In particular, this implies that any power $1/(z - z_1)^k$ can also be approximated uniformly on K by polynomials.

It now suffices to prove that $1/(z - z_0)$ can be approximated uniformly on K by polynomials in $1/(z - z_1)$. To do so, we use the fact that K^c is connected to travel from z_0 to the point z_1. Let γ be a curve in K^c that is parametrized by $\gamma(t)$ on $[0,1]$, and such that $\gamma(0) = z_0$ and $\gamma(1) = z_1$. If we let $\rho = \frac{1}{2}d(K,\gamma)$, then $\rho > 0$ since γ and K are compact. We then choose a sequence of points $\{w_1, \ldots, w_\ell\}$ on γ such that $w_0 = z_0$, $w_\ell = z_1$, and $|w_j - w_{j+1}| < \rho$ for all $0 \le j < \ell$.

We claim that if w is a point on γ, and w' any other point with $|w - w'| < \rho$, then $1/(z - w)$ can be approximated uniformly on K by polynomials in $1/(z - w')$. To see this, note that

$$\begin{aligned}\frac{1}{z - w} &= \frac{1}{z - w'}\frac{1}{1 - \frac{w - w'}{z - w'}} \\ &= \sum_{n=0}^{\infty} \frac{(w - w')^n}{(z - w')^{n+1}},\end{aligned}$$

and since the sum converges uniformly for $z \in K$, the approximation by partial sums proves our claim.

This result allows us to travel from z_0 to z_1 through the finite sequence $\{w_j\}$ to find that $1/(z - z_0)$ can be approximated uniformly on K by polynomials in $1/(z - z_1)$. This concludes the proof of the lemma, and also that of the theorem.

6 Exercises

1. Prove that

$$\int_0^\infty \sin(x^2)\,dx = \int_0^\infty \cos(x^2)\,dx = \frac{\sqrt{2\pi}}{4}.$$

These are the **Fresnel integrals**. Here, $\int_0^\infty$ is interpreted as $\lim_{R\to\infty}\int_0^R$.
[Hint: Integrate the function e^{-z^2} over the path in Figure 14. Recall that $\int_{-\infty}^\infty e^{-x^2}\,dx = \sqrt{\pi}$.]

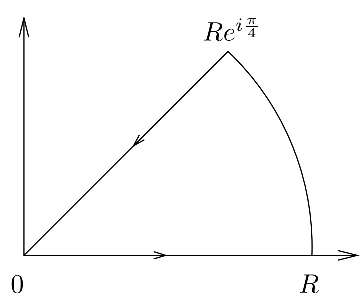

Figure 14. The contour in Exercise 1

2. Show that $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$.
[Hint: The integral equals $\frac{1}{2i}\int_{-\infty}^\infty \frac{e^{ix}-1}{x}\,dx$. Use the indented semicircle.]

3. Evaluate the integrals

$$\int_0^\infty e^{-ax}\cos bx\,dx \qquad \text{and} \qquad \int_0^\infty e^{-ax}\sin bx\,dx\,, \qquad a > 0$$

by integrating e^{-Az}, $A = \sqrt{a^2 + b^2}$, over an appropriate sector with angle ω, with $\cos\omega = a/A$.

4. Prove that for all $\xi \in \mathbb{C}$ we have $e^{-\pi\xi^2} = \displaystyle\int_{-\infty}^{\infty} e^{-\pi x^2} e^{2\pi i x\xi}\, dx$.

5. Suppose f is continuously *complex* differentiable on Ω, and $T \subset \Omega$ is a triangle whose interior is also contained in Ω. Apply Green's theorem to show that

$$\int_T f(z)\, dz = 0.$$

This provides a proof of Goursat's theorem under the additional assumption that f' is continuous.

[Hint: Green's theorem says that if (F, G) is a continuously differentiable vector field, then

$$\int_T F\, dx + G\, dy = \int_{\text{Interior of } T} \left(\frac{\partial G}{\partial x} - \frac{\partial F}{\partial y}\right) dxdy.$$

For appropriate F and G, one can then use the Cauchy-Riemann equations.]

6. Let Ω be an open subset of $\mathbb{C}$ and let $T \subset \Omega$ be a triangle whose interior is also contained in Ω. Suppose that f is a function holomorphic in Ω except possibly at a point w inside T. Prove that if f is bounded near w, then

$$\int_T f(z)\, dz = 0.$$

7. Suppose $f : \mathbb{D} \to \mathbb{C}$ is holomorphic. Show that the diameter $d = \sup_{z,\, w \in \mathbb{D}} |f(z) - f(w)|$ of the image of f satisfies

$$2|f'(0)| \leq d.$$

Moreover, it can be shown that equality holds precisely when f is linear, $f(z) = a_0 + a_1 z$.

Note. In connection with this result, see the relationship between the diameter of a curve and Fourier series described in Problem 1, Chapter 4, Book I.

[Hint: $2f'(0) = \frac{1}{2\pi i}\int_{|\zeta|=r} \frac{f(\zeta)-f(-\zeta)}{\zeta^2}\, d\zeta$ whenever $0 < r < 1$.]

8. If f is a holomorphic function on the strip $-1 < y < 1$, $x \in \mathbb{R}$ with

$$|f(z)| \leq A(1 + |z|)^\eta, \qquad \eta \text{ a fixed real number}$$

for all z in that strip, show that for each integer $n \geq 0$ there exists $A_n \geq 0$ so that

$$|f^{(n)}(x)| \leq A_n(1 + |x|)^\eta, \qquad \text{for all } x \in \mathbb{R}.$$

[Hint: Use the Cauchy inequalities.]

9. Let Ω be a bounded open subset of $\mathbb{C}$, and $\varphi : \Omega \to \Omega$ a holomorphic function. Prove that if there exists a point $z_0 \in \Omega$ such that

$$\varphi(z_0) = z_0 \qquad \text{and} \qquad \varphi'(z_0) = 1$$

then φ is linear.

[Hint: Why can one assume that $z_0 = 0$? Write $\varphi(z) = z + a_n z^n + O(z^{n+1})$ near 0, and prove that if $\varphi_k = \varphi \circ \cdots \circ \varphi$ (where φ appears k times), then $\varphi_k(z) = z + ka_n z^n + O(z^{n+1})$. Apply the Cauchy inequalities and let $k \to \infty$ to conclude the proof. Here we use the standard O notation, where $f(z) = O(g(z))$ as $z \to 0$ means that $|f(z)| \leq C|g(z)|$ for some constant C as $|z| \to 0$.]

10. Weierstrass's theorem states that a continuous function on $[0, 1]$ can be uniformly approximated by polynomials. Can every continuous function on the closed unit disc be approximated uniformly by polynomials in the variable z?

11. Let f be a holomorphic function on the disc D_{R_0} centered at the origin and of radius R_0.

(a) Prove that whenever $0 < R < R_0$ and $|z| < R$, then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \mathrm{Re}\left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z}\right) d\varphi.$$

(b) Show that

$$\mathrm{Re}\left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r}\right) = \frac{R^2 - r^2}{R^2 - 2Rr\cos\gamma + r^2}.$$

[Hint: For the first part, note that if $w = R^2/\overline{z}$, then the integral of $f(\zeta)/(\zeta - w)$ around the circle of radius R centered at the origin is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired identity.]

12. Let u be a real-valued function defined on the unit disc $\mathbb{D}$. Suppose that u is twice continuously differentiable and harmonic, that is,

$$\triangle u(x, y) = 0$$

for all $(x, y) \in \mathbb{D}$.

(a) Prove that there exists a holomorphic function f on the unit disc such that

$$\mathrm{Re}(f) = u.$$

Also show that the imaginary part of f is uniquely defined up to an additive (real) constant. [Hint: From the previous chapter we would have $f'(z) = 2\partial u/\partial z$. Therefore, let $g(z) = 2\partial u/\partial z$ and prove that g is holomorphic. Why can one find F with $F' = g$? Prove that $\mathrm{Re}(F)$ differs from u by a real constant.]

(b) Deduce from this result, and from Exercise 11, the Poisson integral representation formula from the Cauchy integral formula: If u is harmonic in the unit disc and continuous on its closure, then if $z = re^{i\theta}$ one has

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi)\, d\varphi$$

where $P_r(\gamma)$ is the Poisson kernel for the unit disc given by

$$P_r(\gamma) = \frac{1 - r^2}{1 - 2r\cos\gamma + r^2}.$$

13. Suppose f is an analytic function defined everywhere in $\mathbb{C}$ and such that for each $z_0 \in \mathbb{C}$ at least one coefficient in the expansion

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

is equal to 0. Prove that f is a polynomial.
[Hint: Use the fact that $c_n n! = f^{(n)}(z_0)$ and use a countability argument.]

14. Suppose that f is holomorphic in an open set containing the closed unit disc, except for a pole at z_0 on the unit circle. Show that if

$$\sum_{n=0}^{\infty} a_n z^n$$

denotes the power series expansion of f in the open unit disc, then

$$\lim_{n\to\infty} \frac{a_n}{a_{n+1}} = z_0.$$

15. Suppose f is a non-vanishing continuous function on $\overline{\mathbb{D}}$ that is holomorphic in $\mathbb{D}$. Prove that if

$$|f(z)| = 1 \qquad \text{whenever } |z| = 1,$$

then f is constant.
[Hint: Extend f to all of $\mathbb{C}$ by $f(z) = 1/\overline{f(1/\overline{z})}$ whenever $|z| > 1$, and argue as in the Schwarz reflection principle.]

7 Problems

1. Here are some examples of analytic functions on the unit disc that cannot be extended analytically past the unit circle. The following definition is needed. Let

f be a function defined in the unit disc $\mathbb{D}$, with boundary circle C. A point w on C is said to be *regular* for f if there is an open neighborhood U of w and an analytic function g on U, so that $f = g$ on $\mathbb{D} \cap U$. A function f defined on $\mathbb{D}$ cannot be continued analytically past the unit circle if no point of C is regular for f.

(a) Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} \qquad \text{for } |z| < 1.$$

Notice that the radius of convergence of the above series is 1. Show that f cannot be continued analytically past the unit disc. [Hint: Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then $|f(re^{i\theta})| \to \infty$ as $r \to 1$.]

(b) * Fix $0 < \alpha < \infty$. Show that the analytic function f defined by

$$f(z) = \sum_{n=0}^{\infty} 2^{-n\alpha} z^{2^n} \qquad \text{for } |z| < 1$$

extends continuously to the unit circle, but cannot be analytically continued past the unit circle. [Hint: There is a nowhere differentiable function lurking in the background. See Chapter 4 in Book I.]

2.* Let

$$F(z) = \sum_{n=1}^{\infty} d(n) z^n \qquad \text{for } |z| < 1$$

where $d(n)$ denotes the number of divisors of n. Observe that the radius of convergence of this series is 1. Verify the identity

$$\sum_{n=1}^{\infty} d(n) z^n = \sum_{n=1}^{\infty} \frac{z^n}{1 - z^n}.$$

Using this identity, show that if $z = r$ with $0 < r < 1$, then

$$|F(r)| \geq c \frac{1}{1-r} \log(1/(1-r))$$

as $r \to 1$. Similarly, if $\theta = 2\pi p/q$ where p and q are positive integers and $z = re^{i\theta}$, then

$$|F(re^{i\theta})| \geq c_{p/q} \frac{1}{1-r} \log(1/(1-r))$$

as $r \to 1$. Conclude that F cannot be continued analytically past the unit disc.

3. Morera's theorem states that if f is continuous in $\mathbb{C}$, and $\int_T f(z)\, dz = 0$ for all triangles T, then f is holomorphic in $\mathbb{C}$. Naturally, we may ask if the conclusion still holds if we replace triangles by other sets.

(a) Suppose that f is continuous on $\mathbb{C}$, and

$$\int_C f(z)\,dz = 0 \tag{16}$$

for every circle C. Prove that f is holomorphic.

(b) More generally, let Γ be any toy contour, and $\mathcal{F}$ the collection of all translates and dilates of Γ. Show that if f is continuous on $\mathbb{C}$, and

$$\int_\gamma f(z)\,dz = 0 \qquad \text{for all } \gamma \in \mathcal{F}$$

then f is holomorphic. In particular, Morera's theorem holds under the weaker assumption that $\int_T f(z)\,dz = 0$ for all equilateral triangles.

[Hint: As a first step, assume that f is twice real differentiable, and write $f(z) = f(z_0) + a(z - z_0) + b(\overline{z - z_0}) + O(|z - z_0|^2)$ for z near z_0. Integrating this expansion over small circles around z_0 yields $\partial f/\partial\overline{z} = b = 0$ at z_0. Alternatively, suppose only that f is differentiable and apply Green's theorem to conclude that the real and imaginary parts of f satisfy the Cauchy-Riemann equations.

In general, let $\varphi(w) = \varphi(x, y)$ (when $w = x + iy$) denote a smooth function with $0 \le \varphi(w) \le 1$, and $\int_{\mathbb{R}^2} \varphi(w)\,dV(w) = 1$, where $dV(w) = dxdy$, and $\int$ denotes the usual integral of a function of two variables in $\mathbb{R}^2$. For each $\epsilon > 0$, let $\varphi_\epsilon(z) = \epsilon^{-2}\varphi(\epsilon^{-1}z)$, as well as

$$f_\epsilon(z) = \int_{\mathbb{R}^2} f(z - w)\varphi_\epsilon(w)\,dV(w),$$

where the integral denotes the usual integral of functions of two variables, with $dV(w)$ the area element of $\mathbb{R}^2$. Then f_ϵ is smooth, satisfies condition (16), and $f_\epsilon \to f$ uniformly on any compact subset of $\mathbb{C}$.]

4. Prove the converse to Runge's theorem: if K is a compact set whose complement if not connected, then there exists a function f holomorphic in a neighborhood of K which cannot be approximated uniformly by polynomial on K.

[Hint: Pick a point z_0 in a bounded component of K^c, and let $f(z) = 1/(z - z_0)$. If f can be approximated uniformly by polynomials on K, show that there exists a polynomial p such that $|(z - z_0)p(z) - 1| < 1$. Use the maximum modulus principle (Chapter 3) to show that this inequality continues to hold for all z in the component of K^c that contains z_0.]

5.* There exists an entire function F with the following "universal" property: given any entire function h, there is an increasing sequence $\{N_k\}_{k=1}^\infty$ of positive integers, so that

$$\lim_{n\to\infty} F(z + N_k) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$.

(a) Let $p_1, p_2, \ldots$ denote an enumeration of the collection of polynomials whose coefficients have rational real and imaginary parts. Show that it suffices to find an entire function F and an increasing sequence $\{M_n\}$ of positive integers, such that

$$|F(z) - p_n(z - M_n)| < \frac{1}{n} \quad \text{whenever } z \in D_n, \tag{17}$$

where D_n denotes the disc centered at M_n and of radius n. [Hint: Given h entire, there exists a sequence $\{n_k\}$ such that $\lim_{k\to\infty} p_{n_k}(z) = h(z)$ uniformly on every compact subset of $\mathbb{C}$.]

(b) Construct F satisfying (17) as an infinite series

$$F(z) = \sum_{n=1}^{\infty} u_n(z)$$

where $u_n(z) = p_n(z - M_n)e^{-c_n(z-M_n)^2}$, and the quantities $c_n > 0$ and $M_n > 0$ are chosen appropriately with $c_n \to 0$ and $M_n \to \infty$. [Hint: The function e^{-z^2} vanishes rapidly as $|z| \to \infty$ in the sectors $\{|\arg z| < \pi/4 - \delta\}$ and $\{|\pi - \arg z| < \pi/4 - \delta\}$.]

In the same spirit, there exists an alternate "universal" entire function G with the following property: given any entire function h, there is an increasing sequence $\{N_k\}_{k=1}^{\infty}$ of positive integers, so that

$$\lim_{k\to\infty} D^{N_k} G(z) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$. Here $D^j G$ denotes the j^{th} (complex) derivative of G.

3 Meromorphic Functions and the Logarithm

> One knows that the differential calculus, which has contributed so much to the progress of analysis, is founded on the consideration of differential coefficients, that is derivatives of functions. When one attributes an infinitesimal increase ϵ to the variable x, the function $f(x)$ of this variable undergoes in general an infinitesimal increase of which the first term is proportional to ϵ, and the finite coefficient of ϵ of this increase is what is called its differential coefficient... If considering the values of x where $f(x)$ becomes infinite, we add to one of these values designated by x_1, the infinitesimal ϵ, and then develop $f(x_1 + \epsilon)$ in increasing power of the same quantity, the first terms of this development contain negative powers of ϵ; one of these will be the product of $1/\epsilon$ with a finite coefficient, which we will call the residue of the function $f(x)$, relative to the particular value x_1 of the variable x. Residues of this kind present themselves naturally in several branches of algebraic and infinitesimal analysis. Their consideration furnish methods that can be simply used, that apply to a large number of diverse questions, and that give new formulae that would seem to be of interest to mathematicians...
>
> *A. L. Cauchy,* 1826

There is a general principle in the theory, already implicit in Riemann's work, which states that analytic functions are in an essential way characterized by their singularities. That is to say, globally analytic functions are "effectively" determined by their zeros, and meromorphic functions by their zeros and poles. While these assertions cannot be formulated as precise general theorems, there are nevertheless significant instances where this principle applies.

We begin this chapter by considering singularities, in particular the different kind of point singularities ("isolated" singularities) that a holomorphic function can have. In order of increasing severity, these are:

- removable singularities

- poles
- essential singularities.

The first type is harmless since a function can actually be extended to be holomorphic at its removable singularities (hence the name). Near the third type, the function oscillates and may grow faster than any power, and a complete understanding of its behavior is not easy. For the second type the analysis is more straight forward and is connected with the calculus of residues, which arises as follows.

Recall that by Cauchy's theorem a holomorphic function f in an open set which contains a closed curve γ and its interior satisfies

$$\int_\gamma f(z)\,dz = 0.$$

The question that occurs is: what happens if f has a pole in the interior of the curve? To try to answer this question consider the example $f(z) = 1/z$, and recall that if C is a (positively oriented) circle centered at 0, then

$$\int_C \frac{dz}{z} = 2\pi i.$$

This turns out to be the key ingredient in the calculus of residues.

A new aspect appears when we consider indefinite integrals of holomorphic functions that have singularities. As the basic example $f(z) = 1/z$ shows, the resulting "function" (in this case the logarithm) may not be single-valued, and understanding this phenomenon is of importance for a number of subjects. Exploiting this multi-valuedness leads in effect to the "argument principle." We can use this principle to count the number of zeros of a holomorphic function inside a suitable curve. As a simple consequence of this result, we obtain a significant geometric property of holomorphic functions: they are open mappings. From this, the maximum principle, another important feature of holomorphic functions, is an easy step.

In order to turn to the logarithm itself, and come to grips with the precise nature of its multi-valuedness, we introduce the notions of homotopy of curves and simply connected domains. It is on the latter type of open sets that single-valued branches of the logarithm can be defined.

1 Zeros and poles

By definition, a **point singularity** of a function f is a complex number z_0 such that f is defined in a neighborhood of z_0 but not at the point

z_0 itself. We shall also call such points **isolated singularities**. For example, if the function f is defined only on the punctured plane by $f(z) = z$, then the origin is a point singularity. Of course, in that case, the function f can actually be defined at 0 by setting $f(0) = 0$, so that the resulting extension is continuous and in fact entire. (Such points are then called removable singularities.) More interesting is the case of the function $g(z) = 1/z$ defined in the punctured plane. It is clear now that g cannot be defined as a continuous function, much less as a holomorphic function, at the point 0. In fact, $g(z)$ grows to infinity as z approaches 0, and we shall say that the origin is a pole singularity. Finally, the case of the function $h(z) = e^{1/z}$ on the punctured plane shows that removable singularities and poles do not tell the whole story. Indeed, the function $h(z)$ grows indefinitely as z approaches 0 on the positive real line, while h approaches 0 as z goes to 0 on the negative real axis. Finally h oscillates rapidly, yet remains bounded, as z approaches the origin on the imaginary axis.

Since singularities often appear because the denominator of a fraction vanishes, we begin with a local study of the zeros of a holomorphic function.

A complex number z_0 is a **zero** for the holomorphic function f if $f(z_0) = 0$. In particular, analytic continuation shows that the zeros of a non-trivial holomorphic function are isolated. In other words, if f is holomorphic in Ω and $f(z_0) = 0$ for some $z_0 \in \Omega$, then there exists an open neighborhood U of z_0 such that $f(z) \neq 0$ for all $z \in U - \{z_0\}$, unless f is identically zero. We start with a local description of a holomorphic function near a zero.

Theorem 1.1 *Suppose that f is holomorphic in a connected open set Ω, has a zero at a point $z_0 \in \Omega$, and does not vanish identically in Ω. Then there exists a neighborhood $U \subset \Omega$ of z_0, a non-vanishing holomorphic function g on U, and a unique positive integer n such that*

$$f(z) = (z - z_0)^n g(z) \quad \textit{for all } z \in U.$$

Proof. Since Ω is connected and f is not identically zero, we conclude that f is not identically zero in a neighborhood of z_0. In a small disc centered at z_0 the function f has a power series expansion

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k.$$

Since f is not identically zero near z_0, there exists a smallest integer n

such that $a_n \neq 0$. Then, we can write

$$f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z)\,,$$

where g is defined by the series in brackets, and hence is holomorphic, and is nowhere vanishing for all z close to z_0 (since $a_n \neq 0$). To prove the uniqueness of the integer n, suppose that we can also write

$$f(z) = (z - z_0)^n g(z) = (z - z_0)^m h(z)$$

where $h(z_0) \neq 0$. If $m > n$, then we may divide by $(z - z_0)^n$ to see that

$$g(z) = (z - z_0)^{m-n} h(z)$$

and letting $z \to z_0$ yields $g(z_0) = 0$, a contradiction. If $m < n$ a similar argument gives $h(z_0) = 0$, which is also a contradiction. We conclude that $m = n$, thus $h = g$, and the theorem is proved.

In the case of the above theorem, we say that f has a **zero of order** n (or **multiplicity** n) at z_0. If a zero is of order 1, we say that it is **simple**. We observe that, quantitatively, the order describes the rate at which the function vanishes.

The importance of the previous theorem comes from the fact that we can now describe precisely the type of singularity possessed by the function $1/f$ at z_0.

For this purpose, it is now convenient to define a **deleted neighborhood** of z_0 to be an open disc centered at z_0, minus the point z_0, that is, the set

$$\{z : 0 < |z - z_0| < r\}$$

for some $r > 0$. Then, we say that a function f defined in a deleted neighborhood of z_0 has a **pole** at z_0, if the function $1/f$, defined to be zero at z_0, is holomorphic in a full neighborhood of z_0.

Theorem 1.2 *If f has a pole at $z_0 \in \Omega$, then in a neighborhood of that point there exist a non-vanishing holomorphic function h and a unique positive integer n such that*

$$f(z) = (z - z_0)^{-n} h(z).$$

Proof. By the previous theorem we have $1/f(z) = (z - z_0)^n g(z)$, where g is holomorphic and non-vanishing in a neighborhood of z_0, so the result follows with $h(z) = 1/g(z)$.

The integer n is called the **order** (or **multiplicity**) of the pole, and describes the rate at which the function grows near z_0. If the pole is of order 1, we say that it is **simple**.

The next theorem should be reminiscent of power series expansion, except that now we allow terms of negative order, to account for the presence of a pole.

Theorem 1.3 *If f has a pole of order n at z_0, then*

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{(z-z_0)} + G(z)\,, \tag{1}$$

where G is a holomorphic function in a neighborhood of z_0.

Proof. The proof follows from the multiplicative statement in the previous theorem. Indeed, the function h has a power series expansion

$$h(z) = A_0 + A_1(z-z_0) + \cdots$$

so that

$$\begin{aligned} f(z) &= (z-z_0)^{-n}(A_0 + A_1(z-z_0) + \cdots) \\ &= \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{(z-z_0)} + G(z). \end{aligned}$$

The sum

$$\frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{(z-z_0)}$$

is called the **principal part** of f at the pole z_0, and the coefficient a_{-1} is the **residue** of f at that pole. We write $\mathrm{res}_{z_0} f = a_{-1}$. The importance of the residue comes from the fact that all the other terms in the principal part, that is, those of order strictly greater than 1, have primitives in a deleted neighborhood of z_0. Therefore, if $P(z)$ denotes the principal part above and C is any circle centered at z_0, we get

$$\frac{1}{2\pi i}\int_C P(z)\,dz = a_{-1}.$$

We shall return to this important point in the section on the residue formula.

As we shall see, in many cases, the evaluation of integrals reduces to the calculation of residues. In the case when f has a simple pole at z_0, it is clear that

$$\mathrm{res}_{z_0} f = \lim_{z\to z_0}(z-z_0)f(z).$$

If the pole is of higher order, a similar formula holds, one that involves differentiation as well as taking a limit.

Theorem 1.4 *If f has a pole of order n at z_0, then*

$$\mathrm{res}_{z_0} f = \lim_{z\to z_0} \frac{1}{(n-1)!}\left(\frac{d}{dz}\right)^{n-1} (z-z_0)^n f(z).$$

The theorem is an immediate consequence of formula (1), which implies

$$(z-z_0)^n f(z) = a_{-n} + a_{-n+1}(z-z_0) + \cdots + a_{-1}(z-z_0)^{n-1} +$$

$$+G(z)(z-z_0)^n.$$

2 The residue formula

We now discuss the celebrated residue formula. Our approach follows the discussion of Cauchy's theorem in the last chapter: we first consider the case of the circle and its interior the disc, and then explain generalizations to toy contours and their interiors.

Theorem 2.1 *Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C. Then*

$$\int_C f(z)\,dz = 2\pi i \ \mathrm{res}_{z_0} f.$$

Proof. Once again, we may choose a keyhole contour that avoids the pole, and let the width of the corridor go to zero to see that

$$\int_C f(z)\,dz = \int_{C_\epsilon} f(z)\,dz$$

where C_ϵ is the small circle centered at the pole z_0 and of radius ϵ.

Now we observe that

$$\frac{1}{2\pi i}\int_{C_\epsilon} \frac{a_{-1}}{z-z_0}\,dz = a_{-1}$$

is an immediate consequence of the Cauchy integral formula (Theorem 4.1 of the previous chapter), applied to the constant function $f = a_{-1}$. Similarly,

$$\frac{1}{2\pi i}\int_{C_\epsilon} \frac{a_{-k}}{(z-z_0)^k}\,dz = 0$$

when $k > 1$, by using the corresponding formulae for the derivatives (Corollary 4.2 also in the previous chapter). But we know that in a neighborhood of z_0 we can write

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z-z_0} + G(z),$$

where G is holomorphic. By Cauchy's theorem, we also know that $\int_{C_\epsilon} G(z)\,dz = 0$, hence $\int_{C_\epsilon} f(z)\,dz = a_{-1}$. This implies the desired result.

This theorem can be generalized to the case of finitely many poles in the circle, as well as to the case of toy contours.

Corollary 2.2 *Suppose that f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1, \ldots, z_N$ inside C. Then*

$$\int_C f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

For the proof, consider a multiple keyhole which has a loop avoiding each one of the poles. Let the width of the corridors go to zero. In the limit, the integral over the large circle equals a sum of integrals over small circles to which Theorem 2.1 applies.

Corollary 2.3 *Suppose that f is holomorphic in an open set containing a toy contour γ and its interior, except for poles at the points $z_1, \ldots, z_N$ inside γ. Then*

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

In the above, we take γ to have positive orientation.

The proof consists of choosing a keyhole appropriate for the given toy contour, so that, as we have seen previously, we can reduce the situation to integrating over small circles around the poles where Theorem 2.1 applies.

The identity $\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f$ is referred to as the **residue formula**.

2.1 Examples

The calculus of residues provides a powerful technique to compute a wide range of integrals. In the examples we give next, we evaluate three improper Riemann integrals of the form

$$\int_{-\infty}^{\infty} f(x)\,dx.$$

The main idea is to extend f to the complex plane, and then choose a family γ_R of toy contours so that

$$\lim_{R\to\infty}\int_{\gamma_R} f(z)\,dz = \int_{-\infty}^{\infty} f(x)\,dx.$$

By computing the residues of f at its poles, we easily obtain $\int_{\gamma_R} f(z)\,dz$. The challenging part is to choose the contours γ_R, so that the above limit holds. Often, this choice is motivated by the decay behavior of f.

EXAMPLE 1. First, we prove that

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi \tag{2}$$

by using contour integration. Note that if we make the change of variables $x \mapsto x/y$, this yields

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{y\,dx}{y^2+x^2} = \int_{-\infty}^{\infty} \mathcal{P}_y(x)\,dx.$$

In other words, formula (2) says that the integral of the Poisson kernel $\mathcal{P}_y(x)$ is equal to 1 for each $y > 0$. This was proved quite easily in Lemma 2.5 of Chapter 5 in Book I, since $1/(1+x^2)$ is the derivative of $\arctan x$. Here we provide a residue calculation that leads to another proof of (2).

Consider the function

$$f(z) = \frac{1}{1+z^2},$$

which is holomorphic in the complex plane except for simple poles at the points i and $-i$. Also, we choose the contour γ_R shown in Figure 1. The contour consists of the segment $[-R, R]$ on the real axis and of a large half-circle centered at the origin in the upper half-plane.

Since we may write

$$f(z) = \frac{1}{(z-i)(z+i)}$$

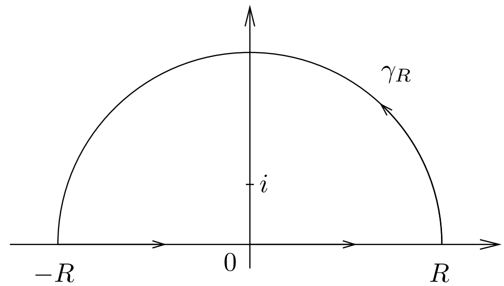

Figure 1. The contour γ_R in Example 1

we see that the residue of f at i is simply $1/2i$. Therefore, if R is large enough, we have

$$\int_{\gamma_R} f(z)\,dz = \frac{2\pi i}{2i} = \pi.$$

If we denote by C_R^+ the large half-circle of radius R, we see that

$$\left| \int_{C_R^+} f(z)\,dz \right| \leq \pi R \frac{B}{R^2} \leq \frac{M}{R},$$

where we have used the fact that $|f(z)| \leq B/|z|^2$ when $z \in C_R^+$ and R is large. So this integral goes to 0 as $R \to \infty$. Therefore, in the limit we find that

$$\int_{-\infty}^{\infty} f(x)\,dx = \pi,$$

as desired. We remark that in this example, there is nothing special about our choice of the semicircle in the upper half-plane. One gets the same conclusion if one uses the semicircle in the lower half-plane, with the other pole and the appropriate residue.

Example 2. An integral that will play an important role in Chapter 6 is

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x}\,dx = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1.$$

To prove this formula, let $f(z) = e^{az}/(1+e^z)$, and consider the contour consisting of a rectangle in the upper half-plane with a side lying

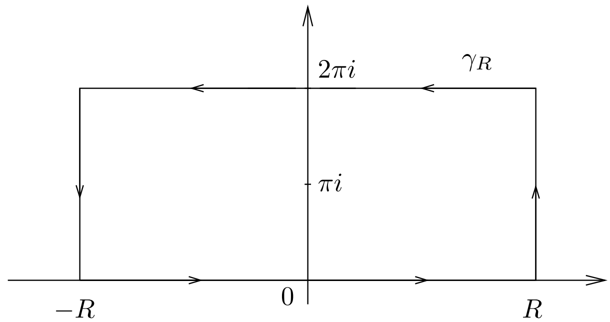

Figure 2. The contour γ_R in Example 2

on the real axis, and a parallel side on the line $\mathrm{Im}(z) = 2\pi$, as shown in Figure 2.

The only point in the rectangle γ_R where the denominator of f vanishes is $z = \pi i$. To compute the residue of f at that point, we argue as follows: First, note

$$(z - \pi i)f(z) = e^{az}\frac{z - \pi i}{1 + e^z} = e^{az}\frac{z - \pi i}{e^z - e^{\pi i}}.$$

We recognize on the right the inverse of a difference quotient, and in fact

$$\lim_{z \to \pi i} \frac{e^z - e^{\pi i}}{z - \pi i} = e^{\pi i} = -1$$

since e^z is its own derivative. Therefore, the function f has a simple pole at πi with residue

$$\mathrm{res}_{\pi i} f = -e^{a\pi i}.$$

As a consequence, the residue formula says that

$$\int_{\gamma_R} f = -2\pi i e^{a\pi i}. \tag{3}$$

We now investigate the integrals of f over each side of the rectangle. Let I_R denote

$$\int_{-R}^{R} f(x)\,dx$$

and I the integral we wish to compute, so that $I_R \to I$ as $R \to \infty$. Then, it is clear that the integral of f over the top side of the rectangle (with the orientation from right to left) is

$$-e^{2\pi i a} I_R.$$

Finally, if $A_R = \{R + it : \; 0 \le t \le 2\pi\}$ denotes the vertical side on the right, then

$$\left| \int_{A_R} f \right| \le \int_0^{2\pi} \left| \frac{e^{a(R+it)}}{1 + e^{R+it}} \right| dt \le Ce^{(a-1)R},$$

and since $a < 1$, this integral tends to 0 as $R \to \infty$. Similarly, the integral over the vertical segment on the left goes to 0, since it can be bounded by Ce^{-aR} and $a > 0$. Therefore, in the limit as R tends to infinity, the identity (3) yields

$$I - e^{2\pi i a} I = -2\pi i e^{a\pi i},$$

from which we deduce

$$\begin{aligned} I &= -2\pi i \, \frac{e^{a\pi i}}{1 - e^{2\pi i a}} \\ &= \frac{2\pi i}{e^{\pi i a} - e^{-\pi i a}} \\ &= \frac{\pi}{\sin \pi a}, \end{aligned}$$

and the computation is complete.

Example 3. Now we calculate another Fourier transform, namely

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh \pi x} \, dx = \frac{1}{\cosh \pi \xi}$$

where

$$\cosh z = \frac{e^z + e^{-z}}{2}.$$

In other words, the function $1/\cosh \pi x$ is its own Fourier transform, a property also shared by $e^{-\pi x^2}$ (see Example 1, Chapter 2). To see this, we use a rectangle γ_R as shown on Figure 3 whose width goes to infinity, but whose height is fixed.

For a fixed $\xi \in \mathbb{R}$, let

$$f(z) = \frac{e^{-2\pi i z \xi}}{\cosh \pi z}$$